

2.1.6

Consider the problem

$$\begin{aligned} & \text{maximize} && x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n} \\ & \text{subject to} && \sum_{i=1}^n x_i = 1, \quad x_i \geq 0, \quad i = 1, \dots, n, \end{aligned}$$

where a_i are given positive scalars. Find a global maximum and show that it is unique.

2.1.12 (Projection on a Simplex)

- (a) Develop an algorithm to find the projection of a vector z on a simplex. This algorithm should be almost as simple as a closed form solution.

2.3.1 (Scaled Gradient Projection Method)

Let H^k be a positive definite matrix and define $\bar{x}^k$ by

$$\bar{x}^k = \arg \min_{x \in X} \left\{ \nabla f(x^k)'(x - x^k) + \frac{1}{2s^k} (x - x^k)' H^k (x - x^k) \right\}.$$

Show that $\bar{x}^k$ solves the problem

$$\begin{aligned} & \text{minimize} && \frac{1}{2} \|x - (x^k - s^k (H^k)^{-1} \nabla f(x^k))\|_{H^k}^2 \\ & \text{subject to} && x \in X, \end{aligned}$$

where $\|\cdot\|_{H^k}$ is the norm defined by $\|z\|_{H^k} = \sqrt{z' H^k z}$.

3.1.1

Use the Lagrange multiplier theorem to solve the following problems:

- (a) $f(x) = \|x\|^2$, $h(x) = \sum_{i=1}^n x_i - 1$.
(b) $f(x) = \sum_{i=1}^n x_i$, $h(x) = \|x\|^2 - 1$.
(c) $f(x) = \|x\|^2$, $h(x) = x' Q x - 1$, where Q is positive definite.